

TOMAS ESTEBAN QUESADA

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EDUCATION

- Pennsylvania State University** State College, PA
Master of Science - Mechanical Engineering; GPA: 3.91
Aug 2023 – Aug 2025
- Pennsylvania State University** State College, PA
Bachelor of Science - Mechanical Engineering; GPA: 3.84
Aug 2018 – May 2023

EXPERIENCE

- Garmin** Olathe, KS
Mechanical Engineer
Aug 2025 – Present
 - Led mechanical design of compact consumer electronic products, owning subsystem architecture, enclosure design, and validation from concept through mass production (125,000+ units).
 - Designed high-density assemblies integrating PCBAs, sensors, and sealed enclosures, including PCB outlines and flex PCB routing, resolving tight tolerance stack-ups and alignment challenges in constrained packaging environments.
 - Collaborated cross-functionally with electrical engineering, industrial design, and manufacturing teams (Taiwan) to optimize injection-molded components for durability, DFM/DFA, and high-volume production.
 - Executed validation testing including thermal shock, drop, and mechanical shock, IP67 waterproof testing, as well as button durability testing exceeding 100,000 cycles.
- Garmin** Olathe, KS
Mechanical Engineering Intern
May 2024 – Aug 2024
 - Designed motor mounts for a live sonar scanner using Onshape and SolidWorks FEA, increasing the factor of safety by 25%.
 - Engineered and validated a hydraulic test bed (1,000 psi, 5,000 lbf) with integrated electronics, enabling automated durability testing of actuator systems up to 1 million cycles.
- Carnegie Robotics** Pittsburgh, PA
Mechanical Engineering Intern
May 2023 – Aug 2023
 - Designed mechanical integration of LiDAR and stereo camera systems on a DARPA-funded autonomous tracked vehicle, including mounts, shielding, and interface constraints.
 - Supported field testing under real-world terrain and impact conditions, validating structural integrity and identifying failure modes in sensor mounts.
 - Designed and fabricated enclosures and mounting systems for autonomous platforms using SolidWorks, supporting multidisciplinary hardware integration.
- Penn State Neuromechanics Laboratory** State College, PA
Graduate Research Assistant
Aug 2023 – Jul 2025
 - Designed and fabricated a tendon-driven soft robotic prosthetic hand with integrated actuation and electronics, focusing on compliant mechanisms and durability.
 - Performed hyperelastic FEA in ANSYS and validated results with a custom-built fatigue and force testing system, achieving a 67.4% reduction in peak stress and 28% lower actuation force over 30,000 cycles.
- GE Appliances** Louisville, KY
Mechanical Engineering Intern
Aug 2021 – Dec 2021
 - Reduced liquid system usage by 30% through nozzle evaluation and testing.
 - Optimized the bill of materials and reduced fastener costs by 40%, improving manufacturability and assembly efficiency.

PROJECTS

- DJI RoboMaster North America Competition** Penn State University
Mechanical Team Lead – Sentry Robot
2024 – 2025
 - Led mechanical design of a competition sentry robot, integrating 5+ subsystems (drive, launcher, gimbal, armor, electronics) under strict weight and size constraints.
 - Engineered a DC motor-driven dual-flywheel launching system for 48mm projectiles, integrating feed mechanisms and structure to increase ammo capacity by 5× while maintaining reliable firing performance.
- Autonomous Maze Navigation Robot** Penn State University
Mechanical & Systems Engineer
2024
 - Designed and fabricated a custom robotic chassis integrating LiDAR, depth camera, and onboard compute, implementing YOLO-based perception and sensor fusion for real-time autonomous navigation.

SKILLS

- CAD Software:** SolidWorks, Onshape, PTC Creo, Fusion 360, Siemens NX (3D mechanical design)
- Simulation & Analysis:** ANSYS, MATLAB, SolidWorks FEA
- Manufacturing:** Injection Molding (DFM/DFA for consumer electronics), CNC, Additive Manufacturing (FDM/PolyJet)
- Testing & Validation:** Consumer product validation (fatigue, drop, shock), Load Cell Calibration, Hydraulic Testing, Waterproofing, Wear and durability testing
- Mechanical Fundamentals:** Tolerance Stack-Up, Fit & Alignment, Load Paths, Wear, Failure Modes, Wearable Product Design

ADDITIONAL

- Awards:** Graduate Research Fellowship, Penn State Provost Award, Capstone Honorable Mention